

input

$$f(n) = \left[\frac{d}{dn} (n^2) \right] \Rightarrow 2n$$

output

$$\begin{array}{ccc} f(1) & \rightarrow & \frac{d}{dn} (1^2) \Rightarrow 2 \\ \downarrow & & \\ f(2) & \rightarrow & \frac{d}{dn} (2^2) \Rightarrow 4 \end{array}$$

Standard Input and output

```
var = input("text_display")
```

└────────── optional
message

→ until unless user enter some input program will never executed

→ input() function will always take input as string

"10" "20"

10 + 20 → 30

"10" + "20" → "1020"

Type Conversions

data type		type conversion function
int	→	int()
float	→	float()
boolean	→	bool()
string	→	str()
list	→	list()
tuple	→	tuple()
set	→	set()
dictionary	→	dict()

`input()` → 10 → "10" → `int("10")` → 10

`a = input()` ← "10"
`num = int(a)` → `int("10")` → 10

`a = input()`

`b = int(a)`

`b = int(input())`

variable = target_type (input())

`a = int(input())`

↓
"10"

`a = int("10")`

↓
10

↖
`a = 10`

Type Conversions :

int $\rightarrow$ int() $\leftarrow$ Integer("10"), boolean, float
string

float $\rightarrow$ float() $\leftarrow$ float string("9.6"), boolean, int

boolean $\rightarrow$ bool() int $\rightarrow$ other than zero $\Rightarrow$ true
zero $\Rightarrow$ False

str $\rightarrow$ " " $\rightarrow$ False
other than " " $\rightarrow$ true

string $\rightarrow$ str() Anything will be converted

`int (bool (""))` → 0

`str (int (bool ("Hi")))` → "1"

`float (int (str (true)))` → ? error

`boolean (false and false)` → ? false

Problems up on Conditional statements and Numbers (Digit Manipulation)

Comparison $\Rightarrow$ $==$, $!=$, $<$, $<=$, $>$, $>=$

Logical $\Rightarrow$ and, or, not

age = 21
gender = "F"

~~F == M~~

$(9 < 10)$ and $(\text{gender} == \text{"M"})$ and $(\text{age} > 20)$

$\downarrow$

T and F

age = 19

$(9 > 1)$ or $(6 > 1)$ and $(\text{age} > 18)$ and $(\text{gender} == \text{"M"})$

$\downarrow$
T

$\downarrow$
T

$\downarrow$
F

T or T or F or T or F and T

T or F or T or F and T

T or T or F and T

T or F and T

T and T

T $\rightarrow$ final value

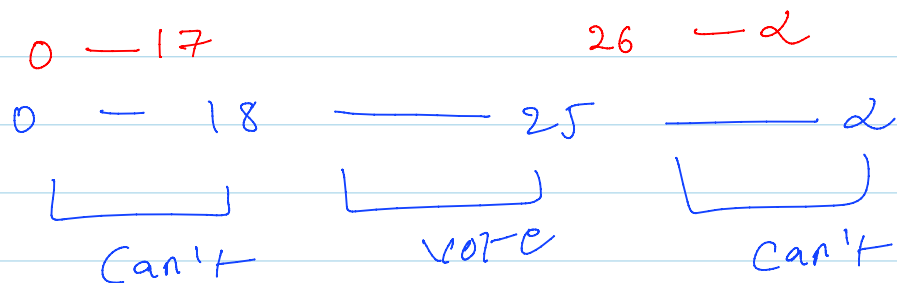
① check weather vote / not ?

② check even / odd / zero

③ check number multiple of 7 or not

if (age >= 18)

if (age > 17)



X if (age < 18 or age > 25) :

Can't
else;

18 < 18 || 18 > 25
↓ ↓
(F || F)

vote ←

if (num == 0)

zero

elif (num % 2 == 0) even

else

odd

equilateral $\rightarrow$ $\begin{matrix} a & b & c \\ 9 & 9 & 9 \end{matrix}$

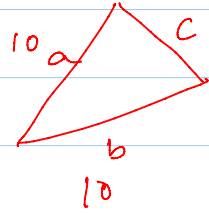
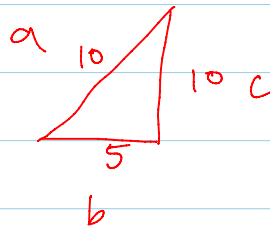
Isosles $\rightarrow$ $\begin{pmatrix} a & b & c \\ 9 & 10 & 9 \end{pmatrix} \begin{pmatrix} a & b & c \\ 10 & 9 & 9 \end{pmatrix} \begin{pmatrix} a & b & c \\ 9 & 9 & 10 \end{pmatrix}$

scalane $\rightarrow$ $\begin{matrix} a & b & c \\ 9 & 10 & 11 \end{matrix}$

$$a = 10$$

$$b = 10$$

$$c = 5$$



$$a = 10$$

$$b = 5$$

$$c = 10$$

if $(a == b \text{ and } b == c \text{ and } c == a)$
equilateral

elif $(a == b \parallel b == c \parallel c == a)$
Isosles

else

Scalane

Math related problems

count notes In a Number 3750 →

$$\begin{array}{r} 500 \times 7 \\ 200 \times 1 \\ 100 \times 0 \\ 50 \times 1 \end{array}$$

$$(500 \times 7 + 200 \times 1 + 100 \times 0 + 50 \times 1)$$

$$1750 - 500 \rightarrow 1250$$

$$1250 - 500 \rightarrow 750$$

$$750 - 500 \rightarrow 250 \leftarrow$$

$$250 - 200 \rightarrow 50$$

$$50 - 50 \rightarrow 0$$

$$\begin{array}{l} \boxed{500} \\ \boxed{200} \end{array}$$

~~100~~

50

$$1750 / 500 \rightarrow 3, 5$$

num = 3750

$$\begin{array}{l} 7 \text{ five} = \text{num} // 500 \leftarrow 7 \times 500 \rightarrow 3500 \\ 250 \text{ num} = \text{num} - \text{five} * 500 \quad 3750 - 3500 \Rightarrow 250 \end{array}$$

$$\begin{array}{l} 1 \text{ two} = \text{num} // 200 \leftarrow 1 \times 200 \rightarrow 200 \\ 50 \text{ num} = \text{num} - \text{two} * 200 \quad 250 - 200 \rightarrow 50 \end{array}$$

$$\begin{array}{l} 0 \text{ one} = \text{num} // 100 \leftarrow 0 \times 100 \rightarrow 0 \\ 50 \text{ num} = \text{num} - \text{one} * 100 \quad 50 - 0 \rightarrow 50 \end{array}$$

$$\begin{array}{l} 1 \text{ fift} = \text{num} // 50 \leftarrow 1 \times 50 \rightarrow 50 \\ 0 \text{ num} = \text{num} - \text{fift} * 50 \quad 50 - 50 \rightarrow 0 \end{array}$$

`print(five)` → 7
`print(two)` → 1
`print(one)` → 0
`print(fift)` → 1

3981

$$7 \times 500 + 2 \times 200 + 0 \times 100 + 1 \times 50 + 1 \times 20 + 1 \times 10 + 1 \times 1$$

Calculate grade according to
Marks $\rightarrow$

marks $\downarrow$

90 - 100 $\rightarrow$ A

80 - 89 $\rightarrow$ B

70 - 79 $\rightarrow$ C

60 - 69 $\rightarrow$ D

50 - 59 $\rightarrow$ E

below 50 $\rightarrow$ F

~~X~~

	T			T	
95	$95 \geq 90$	or	$95 < 100$	$\rightarrow$ T	
	T		F		
120	$120 \geq 90$	or	$120 < 100$	$\rightarrow$ T	

and

if ($m \geq 90$ and $m \leq 100$) { A }

elif ($m \geq 80$ and $m \leq 89$) { B }

⋮

else { F }

$\rightarrow$ This is called checking range

Syntax $\rightarrow$ ($\text{num} \geq \text{start}$ and $\text{num} \leq \text{end}$) ✓

($\text{start} \leq \text{num} \leq \text{end}$) ✗

1 $\leq x \leq 20$

Current Bill Calculation

first 100 - 5
second 100 - 6
third 100 - 7
remaining 100 → d - 8

Input → 3051

1st		2nd		3rd		rest	
100	+	100	+	100	+	5	
100 × 5	+	100 × 6	+	100 × 7	+	5 × 8	
500	+	600	+	700	+	40	→ 1840

output

500 → $100 \times 5 + 100 \times 6 + 100 \times 7 + 200 \times 8$

2000 → $100 \times 5 + 100 \times 6 + 100 \times 7 + 1700 \times 8$

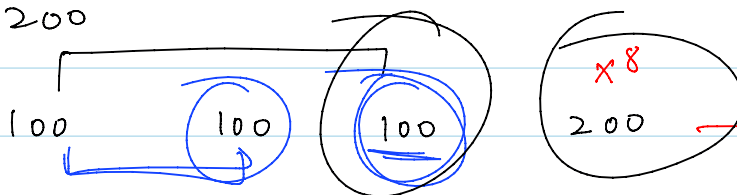
2000 →

$$\begin{array}{r} 100 \times 5 \\ + \\ 100 \times 6 \\ + \\ 100 \times 7 \\ + \\ 100 \\ + \\ 100 \\ + \\ 100 \end{array}$$

x8

$$U = 500 - 300$$

$$\text{Amt} = 0$$



$$500 - 300 = (200 \times 8)$$

2

$$\text{Amt} = (U - 300) \times 8$$

$$U = 300$$

$$\text{Amt} = (U - 200) \times 7$$

$$U = 200$$

$$\text{Amt} = (U - 100) \times 6$$

$$U = 100$$

$$\text{Amt} = (U - 0) \times 5$$

$$u = 150$$

3

$$\text{Amt} = (U - 300) \times 8$$

$$U = 300$$

$$\text{Amt} = (U - 200) \times 7$$

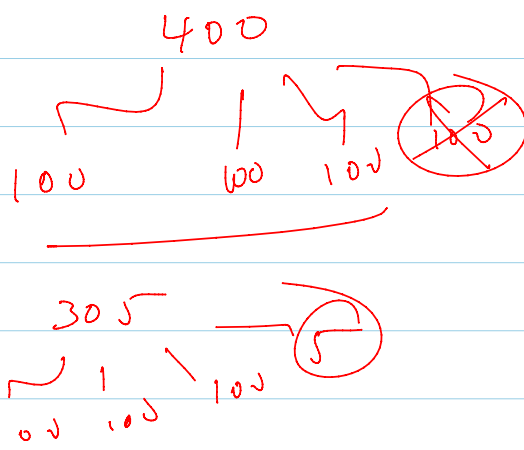
$$U = 200$$

$$\text{Amt} = (U - 100) \times 6$$

$$U = 100$$

$$\text{Amt} = (U - 0) \times 5$$

$$u = 150$$



Bad practice

```
1  amt = 0
2  u = 500
3
4  if(u > 300):
5      amt += (u-300) * 8
6      u = 300
7  elif(u > 200):
8      amt += (u-200) * 7
9      u = 200
10 elif(u > 100):
11     amt += (u-100) * 6
12     u = 100
13
14 amt += (u-0) * 5
15 u = 0
16
17 print(amt)
```

Handwritten annotations for "Bad practice":

- Line 4: $500 > 300 \rightarrow T$
- Line 5: $200 \times 8 \rightarrow 1600$
- Line 6: 300 (with arrow pointing to $u = 300$)
- Line 8: $100 \times 7 \rightarrow 700$
- Line 10: $100 \times 6 \rightarrow 600$
- Line 14: $300 \times 5 \rightarrow 1500$
- Line 15: 0 (with arrow pointing to $u = 0$)
- Final result: $1600 + 1500 + 700 + 600 + 500 = 3900$ (circled in red)

Good practice

```
1  amt = 0
2  u = 500
3
4  if(u > 300):
5      amt += (u-300) * 8
6      u = 300
7  if(u > 200):
8      amt += (u-200) * 7
9      u = 200
10 if(u > 100):
11     amt += (u-100) * 6
12     u = 100
13
14 amt += (u-0) * 5
15 u = 0
16
17 print(amt)
```

Handwritten annotations for "Good practice":

- Line 4: $500 > 300 \rightarrow \text{true}$
- Line 5: $200 \times 8 \rightarrow 1600$
- Line 6: 300 (with arrow pointing to $u = 300$)
- Line 7: $300 > 200 \rightarrow \text{true}$
- Line 8: $100 \times 7 \rightarrow 700$
- Line 9: 200 (with arrow pointing to $u = 200$)
- Line 10: $200 > 100 \rightarrow \text{true}$
- Line 11: $100 \times 6 \rightarrow 600$
- Line 12: 100 (with arrow pointing to $u = 100$)
- Line 14: $100 \times 5 \rightarrow 500$
- Line 15: 0 (with arrow pointing to $u = 0$)
- Final result: $1600 + 700 + 600 + 500 = 3400$ (with arrow pointing to amt)

loops in Python (for, while)

for (i = 0; i < n; i++) { } $\Rightarrow$ C, Java

for i in range(10):

for var in sequence

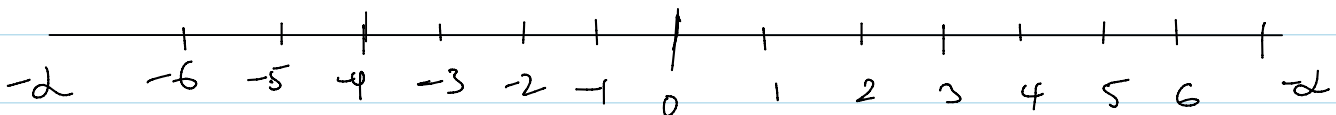
- string
- range (st, end, step)
- collection (list, tuple, set, dict)
- enumerate ()
- zip ()
- zip longest ()

About range() function

syntax: range (start, end, step)

→ here step will play a crucial role

→ step will decide the loop running direction



Syntax: range (start, end, step)

default values:

start $\rightarrow$ default value = 0

end $\rightarrow$ we have to choose (Imp)

step $\rightarrow$ default value = 1

If step +ve $\rightarrow$ val loop check (start < end)
start += step

If step -ve $\rightarrow$ val loop check (start > end)
start += step

Step is the crucial part it will decide whether loop will run or not

Rules

when step $\Rightarrow$ +ve make sure
start < end

when step $\Rightarrow$ -ve make sure
start > end

forward

Backward

for i in range(0, 10, 1)

for i in range(0, -10, -1)

① print from 10 $\rightarrow$ 20 (Inclusive)

② print from 0 $\rightarrow$ -10 (Inclusive)

③ print from 10 $\rightarrow$ -10 (Inclusive)

④ print from -10 $\rightarrow$ 20 (Inclusive)

st < st

① for i in range(10, 21, 1)

st > st

② for i in range(0, -11, -1)

st > stop

③ for i in range(10, -11, -1)

st < stop

④ for i in range(-10, 21, 1)

Nested for loop:-

$\rightarrow$ if we write a loop inside another it will be called nested

for i in range(n):

for j in range(m):

→ for i in range(5) pause

for j in range(5)
print(i)

0	1	2	3	4
0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4

print() function

print(*args, sep=" ", end="\n")

as many
number of
values

separates every
value by
separation

by the end it
will print

```

1 → for i in range(5):
2     for j in range(5):
3         print(i, end=" ")
4     print()

```

0 1 2 3 4

↑

pattern printing

① print a square of $n \times n$

```

1 n = 5
2 for i in range(n):
3     for j in range(n):
4         print("*", end=" ")
5     print()

```

② print a rectangle

$i \rightarrow$
 $col = j$
 $for i \rightarrow 0 \rightarrow 4$
 $for j \rightarrow 0 \rightarrow 5$
 $print(j)$

	0	1	2	3	4	5
0	0	1	2	3	4	5
1	0	1	2	3	4	5
2	0	1	2	3	4	5
3	0	1	2	3	4	5

rows = i

③ left increment triangle

→

↓

rows = i

col = j

0	*				
1	*	*			
2	*	*	*		
3	*	*	*	*	
4	*	*	*	*	*

i value	j - runs for
0	→ 1 ⇒ (i+1) times
1	→ 2 ⇒ (i+1) times
2	→ 3 ⇒ (i+1) times
3	→ 4 ⇒ (i+1) times
4	→ 5 ⇒ (i+1) times

③ Left Decrement

→

↓

rows = i

col = j

0	*	*	*	*	*
1	*	*	*		
2	*	*	*		
3	*	*			
4	*				

i value	j - runs for
0	→ 5 $j = n - i$
1	→ 4 $j = n - i$
2	→ 3 $j = n - i$
3	→ 2 $j = n - i$
4	→ 1 $j = n - i$

f(x)

x →

→ val

0 →

→ 5

1 →

→ 4